

Creation of a Redshift Cluster

Screenshots of the configuration of the Redshift cluster that you have created:

Amazon Redshift > Clusters > redshift-cluster-2

redshift-cluster-2

Actions Edit Add partner integration Query data

General information Info			
Cluster identifier redshift-cluster-2	Status Available	Node type dc2.large	Endpoint redshift-cluster-2.cfuxtifva8sl.us-east-1.redshift.amazonaws.com:5439/dev
Custom domain name -	Date created February 11, 2025, 19:12 (UTC+05:30)	Number of nodes 2	JDBC URL jdbc:redshift://redshift-cluster-2.cfuxtifva8sl.us-east-1.redshift.amazonaws.com:5439/dev
Cluster namespace ARN arn:aws:redshift:us-east-1:905418235757:namespace:271a5058-a229-405e-bb18-aaa42066d6c1	Multi-AZ No	Patch version Patch 187	ODBC URL Driver={Amazon Redshift (x64)}; Server=redshift-cluster-2.cfuxtifva8sl.us-east-1.redshift.amazonaws.com; Database=dev

Queries used for creating the Dimension and Fact tables on the Redshift cluster along with screenshots of the successful status of the query

Setting up a database in the Redshift cluster and running queries to create the dimension and fact tables

Queries to create the various dimension and fact tables with appropriate primary and foreign keys:

creating DIM_LOCATION table

```
create table if not exists etl_atm_data.DIM_LOCATION(
location_id int not null sortkey distkey,
location varchar(50),
streetname varchar(255),
street_number int,
zipcode int,
```

```
lat decimal(10,3),  
lon decimal(10,3),  
PRIMARY KEY(location_id)  
);
```

creating DIM_ATM table

```
create table if not exists etl_atm_data.DIM_ATM(  
atm_id int not null distkey sortkey,  
atm_number varchar(20),  
atm_manufacturer varchar(50),  
atm_location_id int,  
PRIMARY KEY(atm_id),  
FOREIGN KEY(atm_location_id) REFERENCES  
etl_atm_data.DIM_LOCATION(location_id)  
);
```

creating DIM_DATE table

```
create table if not exists etl_atm_data.DIM_DATE(  
date_id int not null distkey sortkey,  
full_date_time timestamp,  
year int,  
month varchar(20),  
day int,  
hour int,  
weekday varchar(20),  
PRIMARY KEY(date_id)  
);
```

creating DIM_CARD_TYPE table

```
create table if not exists etl_atm_data.DIM_CARD_TYPE(  
card_type_id int not null distkey sortkey,  
card_type varchar(30),  
PRIMARY KEY(card_type_id)  
);
```

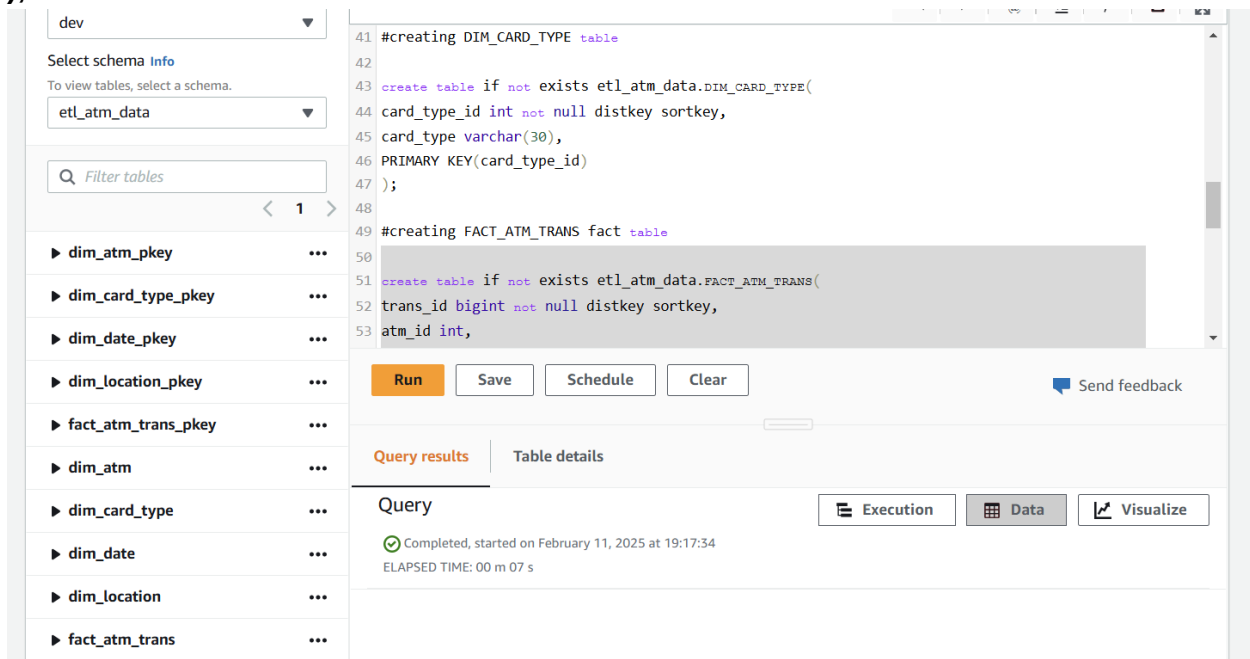
#creating FACT_ATM_TRANS fact table

```
create table if not exists etl_atm_data.FACT_ATM_TRANS(  
trans_id bigint not null distkey sortkey,
```

```

atm_id int,
weather_loc_id int,
date_id int,
card_type_id int,
atm_status varchar(20),
currency varchar(10),
service varchar(20),
transaction_amount int,
message_code varchar(255),
message_text varchar(255),
rain_3h decimal(10,3),
clouds_all int,
weather_id int,
weather_main varchar(50),
weather_description varchar(255),
PRIMARY KEY(trans_id),
FOREIGN KEY(weather_loc_id) REFERENCES
etl_atm_data.DIM_LOCATION(location_id),
FOREIGN KEY(atm_id) REFERENCES etl_atm_data.DIM_ATM(atm_id),
FOREIGN KEY(date_id) REFERENCES etl_atm_data.DIM_DATE(date_id),
FOREIGN KEY(card_type_id) REFERENCES
etl_atm_data.DIM_CARD_TYPE(card_type_id)
);

```



The screenshot shows a SQL query editor interface. On the left, there is a sidebar with a schema browser showing a list of tables: dim_atm_pkey, dim_card_type_pkey, dim_date_pkey, dim_location_pkey, fact_atm_trans_pkey, dim_atm, dim_card_type, dim_date, dim_location, and fact_atm_trans. The main area displays the SQL query for creating two tables: DIM_CARD_TYPE and FACT_ATM_TRANS. The query is as follows:

```

41 #creating DIM_CARD_TYPE table
42
43 create table if not exists etl_atm_data.DIM_CARD_TYPE(
44   card_type_id int not null distkey sortkey,
45   card_type varchar(30),
46   PRIMARY KEY(card_type_id)
47 );
48
49 #creating FACT_ATM_TRANS fact table
50
51 create table if not exists etl_atm_data.FACT_ATM_TRANS(
52   trans_id bigint not null distkey sortkey,
53   atm_id int,

```

Below the query, there are buttons for 'Run', 'Save', 'Schedule', and 'Clear'. A 'Send feedback' link is also present. The interface shows the query has been executed successfully, with a status message: 'Completed, started on February 11, 2025 at 19:17:34 ELAPSED TIME: 00 m 07 s'. There are also tabs for 'Query results', 'Table details', 'Execution', 'Data', and 'Visualize'.

Loading data into a Redshift cluster from Amazon S3 bucket

Queries to copy the data from S3 buckets to the Redshift cluster in the appropriate tables

#loading data into DIM_LOCATION table

```
copy etl_atm_data.DIM_LOCATION from 's3://aws-logs-905418235757-us-east-1/elasticmapreduce/j-1E6SEFG9GA4YQ/etl/dim_location/part-00000-4497a930-8930-45db-93a5-705c7d58d299-c000.csv' iam_role 'arn:aws:iam::905418235757:role/myRedshiftRole' delimiter ',' region 'us-east-1' CSV;
```

#loading data into DIM_ATM table

```
copy etl_atm_data.DIM_ATM from 's3://aws-logs-905418235757-us-east-1/elasticmapreduce/j-1E6SEFG9GA4YQ/etl/dim_atm/part-00000-dc0c784f-40a6-4330-a08d-553d23e030a5-c000.csv' iam_role 'arn:aws:iam::905418235757:role/myRedshiftRole' delimiter ',' region 'us-east-1' CSV;
```

#loading data into DIM_DATE table

```
copy etl_atm_data.DIM_DATE from 's3://aws-logs-905418235757-us-east-1/elasticmapreduce/j-1E6SEFG9GA4YQ/etl/dim_date/part-00000-339f7738-7b0f-47ba-8fed-c81f7c1579e2-c000.csv' iam_role 'arn:aws:iam::905418235757:role/myRedshiftRole' delimiter ',' region 'us-east-1' CSV
timeformat
'auto';
```

#loading data into DIM_CARD_TYPE table

```
copy etl_atm_data.DIM_CARD_TYPE from 's3://aws-logs-905418235757-us-east-1/elasticmapreduce/j-1E6SEFG9GA4YQ/etl/dim_card_type/part-00000-666ce7dc-4bc4-4312-99db-c75d5c570b8f-c000.csv' iam_role 'arn:aws:iam::905418235757:role/myRedshiftRole' delimiter ',' region 'us-east-1' CSV;
```

#loading data into FACT_ATM_TRANS table

```
copy etl_atm_data.FACT_ATM_TRANS from 's3://aws-logs-905418235757-us-east-1/elasticmapreduce/j-1E6SEFG9GA4YQ/etl/fact_atm_trans/part-00000-f625ca03-24bb-4275-9a39-fb54a1eeb281-c000.csv' iam_role 'arn:aws:iam::905418235757:role/myRedshiftRole' delimiter ',' region 'us-east-1' CSV;
```